A Charge + Light Self-Supervised Foundation Model for MicroBooNE LArTPC Open Data

Technical Note — Round 1

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Abstract

We present a prototype multi-modal foundation model for liquid-argon TPC data that takes reconstructed charge hits and PMT light as a single token sequence. A Transformer backbone is pretrained without labels, by masked autoencoding, on 831k MicroBooNE overlay events. We then evaluate it on four downstream tasks with small task heads: hit-level and event-level neutrino/cosmic separation, flavor classification (ν_μCC/ν_eCC/NC) and neutrino-energy regression. Labeled-data fractions range from 0.1% to 100%.

Pretraining clearly helps when labels are scarce. With 0.1% labels, the fine-tuned charge-plus-light model beats training from scratch on every task; for example, hit-level F1 is 0.77 versus 0.22.

Adding light improves every task and is decisive for neutrino/cosmic separation: event-level AUC is 0.97 versus 0.65 for charge only.

At fixed compute, a 6.6M-parameter model saturates at about 10% of the pretraining data. A 21M-parameter model trained twice as long improves all probes, and with it more pretraining data helps consistently. On neutrino/cosmic separation the best models clearly beat the Pandora and flash-PE reference baselines: hit F1 0.95 versus 0.63, and event AUC 0.999 versus 0.82 and 0.91.

We also show that, with fixed-grid charge patches, the masked-reconstruction target is not learnable unless each patch is positioned at its charge centroid. Round 1 used about 29 Polaris node-hours.

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1 Introduction and goals

Foundation models (FMs) learn reusable representations from unlabeled data. Small task-specific heads are then trained on top, as in language and vision. Recent neutrino and nuclear-physics work applies this idea to 3D charge point clouds [6, 9, 10] and to collider TPC space-points [8].

A LArTPC records two correlated measurements of the same event:

- ionization charge on the wire planes;
- scintillation light on the PMTs.

Classical reconstruction relates the two explicitly, for example through Wire-Cell's many-to-many charge–light matching [2, 3]. This round tests whether a single self-supervised backbone can learn these correlations, and the physics they carry, directly from paired charge and light.

Round 1 asks three questions, using the MicroBooNE public open samples [1]:

Q1 Does SSL pretraining help? We compare a downstream model trained from scratch with a frozen FM plus head and with a fine-tuned FM, across label budgets.

Q2 Does light help? We compare charge-only (C) with charge+light (CL) inputs.

Q3 How does the FM scale with the amount of pretraining data (1%, 10%, 50%, 90%)?

The compute allowance was 100 node-hours on the Polaris `debug`/`debug-scaling` queues, which have a 1-hour walltime. Section 6 gives the stopping criterion.

2 Data and tasks

2.1 Samples, preprocessing and splits

We use two NoWire HDF5 samples:

- the BNB inclusive overlay (Zenodo 8370883): 753,467 events, 195 GB;
- the ν_e-CC overlay (Zenodo 7261921): 18 of 20 files, 77,340 events.

Together they give 830,807 events. Both samples are GENIE+Geant4 neutrinos overlaid on real off-beam cosmic data, so cosmic hits carry no MC truth.

The HDF5 tables are converted into a column-wise, memory-mappable cache (146 GB for the inclusive sample):

- **per hit:** plane, wire, time, integral, rms; truth (deposited neutrino energy and category); and a Pandora ν-slice flag, used only as a baseline;
- **per optical hit:** channel, time and PE;
- **per event:** truth, the Pandora score and the largest beam flash.

Splits are assigned by a hash of (run, subrun). Overlays that share an off-beam cosmic frame therefore stay in one partition. The pretraining subsets (1/10/50/90% of all events) and the label pools (0.1/1/10/100% of train) are nested random subsets of the train partition.

Table 1: Partitions of the merged corpus. Following the review decision, the ν_e sample is mixed into every partition.

Split	Events	ν_e-sample share
train	747,068	9.4%
val	41,912	9.2%
test	41,827	7.2%

Audit (inclusive sample).

- 15.4k hits per event (median).
- 44% of events have the ν vertex in the active volume (AV).
- In 21% of events the neutrino leaves ≤ 5 hits.
- The beam flash follows the neutrino interaction time by 0.478 μs (5–95%: 0.457–0.507 μs).
- About 1% of optical hits fall outside the 23.4 μs beam window and are dropped.

2.2 Downstream tasks

Table 2 summarizes the tasks. All definitions live in `qlfm/data/labels.py`.

Table 2: Downstream task definitions. AV = active volume; FV = fiducial volume, 3 cm inside the AV; "visible" means ≥ 30 collection-plane ν hits.

Task	Definition	Train counts	Metric
T1 hit ν/cosmic	Per hit: ν if the MC ν-deposit exceeds 0.05 MeV. Evaluated on events with a visible in-AV ν	$\sim$6% ν hits	charge-wtd F1
T2 event ν/cosmic	0 = no ν charge ($\leq$5 ν hits); 1 = visible in-AV ν; 2 = out-of-AV ν leaving >5 hits	142k/333k/241k	AUC (1 vs 0)
T3 flavor	FV + visible: ν_μCC / ν_eCC / NC	200k/68k/49k	macro-F1
T4 energy	FV + visible + CC; regress $\log E_\nu$	268k	resolution†

† Half the 16–84% interval of $(E_{\text{pred}} - E_{\text{true}})/E_{\text{true}}$.

Reference baselines.

- **T1:** Pandora hits in the slice with the highest `nu_score`.
- **T2:** Pandora's maximum `nu_score` gives AUC 0.82; the beam-flash PE in 3–6 μs gives AUC 0.91.

3 Model

Figure 1 shows the architecture.

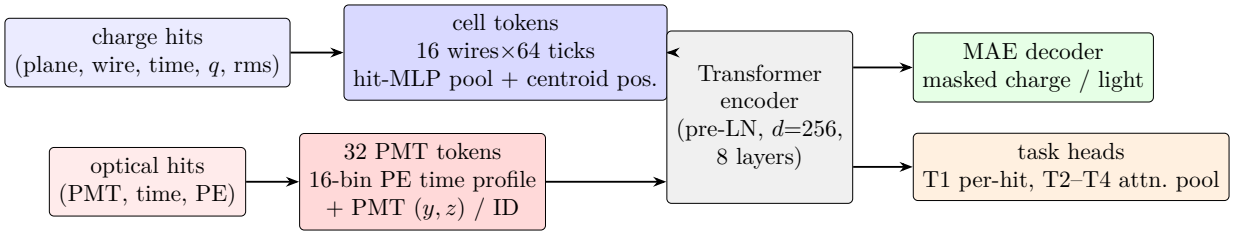


Figure 1: Architecture. Charge cells and PMTs become 256-dimensional tokens in a shared space, with a modality embedding added to each. Pretraining uses the MAE decoder; downstream tasks attach small heads to the encoder.

3.1 Tokenization

Charge tokens.

- Each wire plane is divided into 16-wire × 64-tick cells, about $4.8 \times 3.5\,\mathrm{cm}^2$. Every occupied cell becomes one token.
- This gives about 2.0k tokens per event (median; 99th percentile 3.4k; capped at 4096).
- A shared per-hit MLP takes each hit's relative wire, relative time, $\log q$ and rms. Its outputs are max- and mean-pooled over the cell and concatenated with the cell's hit count and total charge.
- The token's position is its charge-weighted centroid in physical units (drift x and wire coordinate, in cm). It is encoded by 10-octave Fourier features, plus an embedding of the centroid's offset inside the cell, plus a plane embedding.

Light tokens.

- There are 32 tokens per event, one per PMT.
- Each carries $\log(1 + \mathrm{PE})$ in 16 time bins spanning 0–24 μs, binned more finely around the expected neutrino flash, plus the total PE.
- The position is encoded by Fourier features of the PMT (y, z) plus a PMT-ID embedding.

A modality embedding is added to every token. The charge-only model simply has no light tokens. Tokenization runs on the GPU from raw hits.

3.2 Backbone

The encoder is a pre-LN Transformer [13]: 8 layers, token width d=256, 8 heads of dimension 32, and MLP hidden size 1024. It has 6.6M parameters. Pre-LN applies LayerNorm to the input of each sublayer and leaves the residual path untouched:

$$x \leftarrow x + \mathrm{Attn}(\mathrm{LN}(x)), \qquad x \leftarrow x + \mathrm{MLP}(\mathrm{LN}(x)).$$

This trains more stably than the original post-LN form. Downstream heads read the encoder output without a final LayerNorm, following [6]. Events are packed into variable-length sequences and processed with FlashAttention-2 [14].

3.3 Pretraining objective

We use a multi-modal masked autoencoder [4, 7].

Charge.

- 60% of the charge tokens are masked.
- Half of the masked budget comes from 256-tick drift-time strips masked jointly on all three planes, so the model cannot copy a hidden region from another plane's view.
- The decoder (d=192, 3 layers) predicts, for each masked cell, a 4×8 in-cell charge sub-grid plus the cell's log hit count and log total charge. Each target dimension is standardized per batch.

Light. Each event uses one of three masking modes:

- 50% of the PMTs are masked, with probability 0.4;
- all 32 PMTs are masked, with probability 0.4. This makes the task *charge→light* prediction, a learned stand-in for a photon library combined with flash matching;
- nothing is masked, with probability 0.2.

Schedule. Training runs for 17k steps at a global batch of 256 events, on one Polaris node (4×A100) in about 45 min. We use AdamW with learning rate 1.5×10^{-4}, 5% warmup, cosine decay and bf16. The step count is the same for every pretraining fraction.

3.4 A fixed-grid MAE target needs patch centres

In the first version, each charge token was positioned at its cell centre. The masked-charge loss never beat a per-dimension mean predictor, while the light reconstruction learned normally.

Diagnosis.

- A single-batch overfit test and a target-leak test confirmed that the code was wired correctly.
- Fourier, learned-grid and combined position encodings all failed in the same way.
- A non-neural baseline settled the question. Predicting a masked cell's (hit count, total charge) as the average over its visible neighbours is *worse* than predicting the global mean, for 8×32, 16×64 and 32×128 cells.

Tracks cross a fixed grid at arbitrary offsets: a corner clip gives 1–2 hits, while a central crossing gives about 16. Unless the model knows where the track passes through the cell, the cell's content is not predictable.

Fix. Following Point-MAE [5] and PoLAr-MAE [6], tokens are positioned at their charge centroid, together with the centroid's offset inside the cell. With this change the held-out masked-charge error drops from 0.97–0.99 to 0.61× the mean baseline.

3.5 Heads and adaptation modes

Heads.

- **Event-level tasks (T2–T4):** an attentive-pooling head, in which 4 learned queries cross-attend to all tokens, followed by an MLP. It has about 0.5M parameters.
- **Hit-level task (T1):** a per-hit MLP that takes the hit embedding concatenated with the contextual feature of its token.

Modes.

- *scratch*: random initialization; everything is trained.
- *frozen*: the backbone is fixed and only the head is trained.
- *FT*: everything is trained, with layer-wise learning-rate decay 0.75.

Training and evaluation.

- Each run uses one GPU, batch size 32, and clip(30 epochs, 400, 15000) steps.
- Early stopping uses a random validation subset.
- Tests use a random test subset of up to 20k events (3k for T1).
- Label budgets $\leq$1% are run with three seeds.

4 Experiments

Table 3: Experiment overview. Total compute was about 29 node-hours (`sbank`: 29.6 charged on suballocation `debug`). Jobs were submitted by a queue-aware feeder; several single-GPU runs shared each node through a lock-based task queue.

ID	Content	Runs
E1	Scratch sanity checks of the label pipelines (inclusive-only corpus)	4
E2	Pretraining of FM-C and FM-CL on 90% of the data	2
E3	{scratch, frozen, FT} $\times$ {C, CL} $\times$ {0.1, 1, 10, 100}% labels $\times$ 4 tasks; 3 seeds at $\leq$1%	192
E4	Pretraining of FM-C and FM-CL on 1/10/50% of the data; frozen probes at 1% and 10% labels	6 + 48
E5	Checks: learning-rate scan for scratch-C; layer decay 0.9 for hit fine-tuning	8
E6	Larger model (21M parameters, 2$\times$ steps), FM-CL on 10% and 90%; frozen and FT probes	2 + 24

5 Results

5.1 Label efficiency and the value of light (E3)

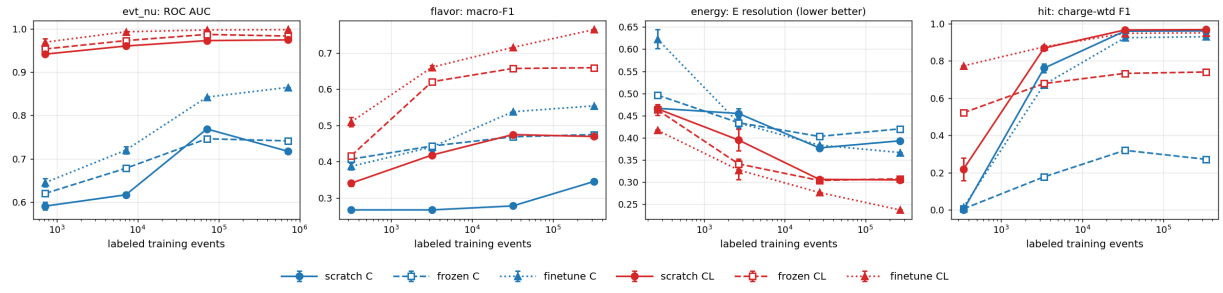


Figure 2: Test metrics versus the number of labeled training events. Colour: input (blue C, red CL). Marker and line: mode (scratch $\circ$ solid, frozen $\square$ dashed, FT $\triangle$ dotted). Error bars show the seed spread at $\leq$1% labels.

Table 4: E3 test metrics at 0.1 / 1 / 10 / 100% labels. Entries at ≤1% are the mean of three seeds; the seed spread is typically ≤ 0.01. For T4, lower is better.

Task (metric)	Mode	C	CL
T2 event ν (AUC)	scratch	0.59 / 0.62 / 0.77 / 0.72	0.94 / 0.96 / 0.97 / 0.98
	frozen	0.62 / 0.68 / 0.75 / 0.74	0.95 / 0.97 / 0.99 / 0.98
	FT	0.65 / 0.72 / 0.84 / 0.87	**0.97 / 0.99 / 1.00 / 1.00**
T3 flavor (macro-F1)	scratch	0.27 / 0.27 / 0.28 / 0.35	0.34 / 0.42 / 0.48 / 0.47
	frozen	0.41 / 0.44 / 0.47 / 0.47	0.42 / 0.62 / 0.66 / 0.66
	FT	0.39 / 0.44 / 0.54 / 0.55	**0.51 / 0.66 / 0.72 / 0.77**
T4 energy (resolution)	scratch	0.47 / 0.46 / 0.38 / 0.39	0.47 / 0.40 / 0.31 / 0.31
	frozen	0.50 / 0.43 / 0.40 / 0.42	0.46 / 0.34 / 0.30 / 0.31
	FT	0.62 / 0.43 / 0.38 / 0.37	**0.42 / 0.33 / 0.28 / 0.24**
T1 hit (charge-wtd F1)	scratch	0.00 / 0.76 / 0.96 / 0.96	0.22 / 0.87 / **0.97 / 0.97**
	frozen	0.01 / 0.18 / 0.32 / 0.27	0.52 / 0.68 / 0.73 / 0.74
	FT	0.01 / 0.67 / 0.92 / 0.93	**0.77 / 0.88** / 0.95 / 0.95

Best model (FT-CL, 100% labels).

- **T2:** AUC 0.999. Signal efficiency is 0.997 at 95% background rejection and 0.973 at 99%. The in-AV versus out-of-AV AUC is 0.96, and the 3-class macro-F1 is 0.82.
- **T3:** ν_μCC efficiency/purity 0.88/0.92; ν_eCC 0.80/0.72 (AUC 0.96); NC 0.65/0.63.
- **T4:** resolution 23.7%, bias −2.5%.
- **T1:** charge-weighted efficiency/purity 0.948/0.953. Pandora reaches 0.78/0.56 on the same events.

Interpretation.

- **Q1: pretraining helps most with few labels.** The advantage shrinks as labels grow.
  - On T1, scratch catches up by 1% labels and is slightly ahead at ≥10%.
  - On the event-level tasks, FT-CL stays ahead even at 100%. Scratch is limited there by the downstream budget (≤15k steps, less than one epoch), so part of that gap is undertraining.
- **The frozen CL backbone is a strong feature extractor for event-level tasks.** It matches or beats scratch at every budget on T2–T4.
- **Frozen features are weak for per-hit labels.** The per-hit embedding receives no direct pretraining signal.
- **Q2: light dominates neutrino/cosmic separation.**
  - Charge-only models must find a small interaction among roughly 26 cosmic muons.
  - The in-time PMT pattern points directly to the neutrino. This is the learned counterpart of charge–light matching.
  - Light also improves flavor and energy.
- **Charge-only scratch fails on event-level tasks** at every tested learning rate (E5). Fine-tuning the same architecture from the C foundation model succeeds: flavor 0.54–0.55 and T2 AUC 0.84–0.87 at 10–100% labels.
- **No sample-origin shortcut is seen for flavor.** The ν_eCC efficiency on inclusive-sample ν_e events (n=79) is equal to or higher than on ν_e-sample events: 0.86 versus 0.80 for FT-CL.

5.2 Pretraining-data scaling (E4, E6)

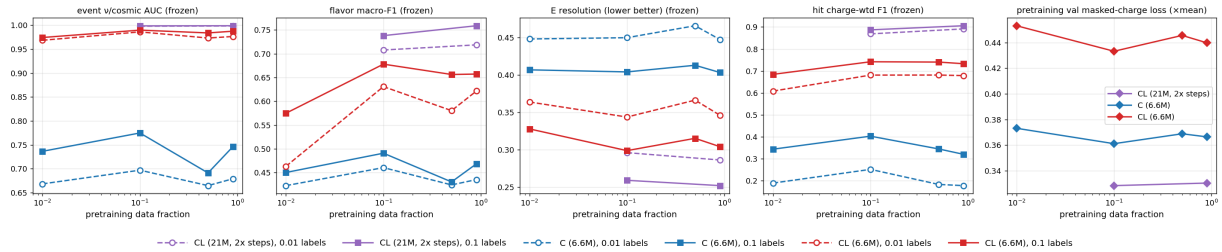


Figure 3: Frozen-probe test metrics and pretraining validation masked-charge loss versus the pretraining-data fraction. Red and blue: 6.6M CL and C models at fixed compute (17k steps). Purple: 21M CL model at 2× steps. Open circles with dashed lines: 1% labels; filled squares with solid lines: 10% labels.

Small model at fixed compute (E4).

- CL improves from 1% to 10% of the pretraining data. At 1% labels: flavor 0.46 → 0.63, T1 0.61 → 0.68, T4 0.36 → 0.34, T2 0.968 → 0.986.
- Above 10% the probes are flat within single-seed jitter of ±0.02–0.04. C is flat throughout.
- The pretraining validation loss has the same shape: CL 0.453, 0.433, 0.446, 0.440; C 0.373, 0.361, 0.369, 0.366.
- Every run sees the same 4.4M events. With 10% of the data (83k events, about 53 epochs), the 6.6M-parameter model already saturates.

Larger model (E6). We pretrain a CL model with d=384, 12 layers and 6 heads (21M parameters) for 34k steps (about 8.7M events) on 10% and on 90% of the data.

Table 5: Pretraining validation loss, in units of the mean-predictor baseline.

Model	Data	Masked charge	Masked light	Charge→light
S (6.6M)	10%	0.433	0.448	0.559
S (6.6M)	90%	0.440	0.446	0.556
M (21M, 2× steps)	10%	0.328	0.431	0.539
M (21M, 2× steps)	90%	0.330	0.413	0.512

Table 6: CL probe test metrics for the small (S) and large (M) models at 10% and 90% pretraining data. Bold marks where the 90% model beats the 10% model.

Probe	S 10%	S 90%	M 10%	M 90%
T2 AUC, frozen, 1% / 10% labels	0.986 / 0.990	0.976 / 0.987	0.998 / 0.999	0.998 / 0.999
T2 AUC, FT, 1% labels	—	0.993	0.998	0.999
T3 macro-F1, frozen, 1% / 10%	0.631 / 0.678	0.622 / 0.657	0.708 / 0.738	**0.719 / 0.759**
T3 macro-F1, FT, 1%	—	0.663	0.724	**0.741**
T4 resolution, frozen, 1% / 10%	0.344 / 0.299	0.346 / 0.304	0.296 / 0.259	**0.286 / 0.252**
T4 resolution, FT, 1%	—	0.307	0.292	**0.274**
T1 F1, frozen, 1% / 10%	0.681 / 0.742	0.678 / 0.733	0.869 / 0.887	**0.891 / 0.906**
T1 F1, FT, 1%	—	0.875	0.925	**0.943**

What E6 shows.

- **Capacity and compute dominate at this scale.** Going from S to M improves every probe: frozen T1 F1 by +0.15 to +0.19, frozen T3 macro-F1 by +0.08 to +0.10, and T4 resolution by about −0.05.
- **Frozen M features rival a fully trained small model.** At 10% labels, frozen M reaches T3 0.759 and T4 0.252. The small model fine-tuned on all labels reaches 0.765 and 0.237.
- **More pretraining data now helps.** With M, 90% beats 10% in all 9 non-saturated comparisons (T3 +0.011 to +0.021; T4 −0.007 to −0.018; T1 +0.018 to +0.022). With S, the same comparisons were flat or reversed.
- **The data benefit shows up in the light objectives.** Charge→light improves from 0.539 to 0.512, while the masked-charge loss is unchanged. The extra data mainly sharpens the learned charge–light correspondence.
- These are single seeds. The evidence is the consistent sign across all 9 comparisons, not the size of any single difference.

5.3 Checks (E5)

- **Scratch-C learning rate.** At 10% labels, learning rates 10^{-4} and 10^{-3} give flavor F1 0.277 and 0.267 and T2 AUC 0.69 and 0.56. The default 3×10^{-4} gives 0.277 and 0.77. The scratch failure is not a learning-rate artifact.
- **Hit fine-tuning with layer decay 0.9** (default 0.75).
  - At 10% labels: C 0.952 (was 0.924) and CL 0.960 (was 0.948), level with scratch (0.959/0.966).
  - At 0.1% labels: CL 0.788 (was 0.773).
  - A decay of 0.9 is recommended for dense tasks.

6 Conclusion of round 1

The round was to stop once the performance was decent and the three comparisons made physical sense, or once further iteration stopped improving the results. That point has been reached:

1. Pretraining clearly beats scratch in the low-label regime, and the advantage narrows as labels grow. For per-hit labels it disappears by about 1% labels, once fine-tuning uses a sensible layer decay.
2. Charge+light is better than charge only on every task, and decisively so for neutrino/cosmic separation. The learned charge→light prediction is non-trivial.
3. Pretraining-data scaling is limited by compute and capacity for the 6.6M model. It reappears consistently with a larger model trained for longer.
4. On neutrino/cosmic separation, the best models clearly beat the Pandora and flash-PE reference baselines.

Further gains are expected from scale, from the pretraining objective and from hit-level design. These are round-2 topics (Section 8).

7 Limitations

- **Overlay data only.** There is no beam-on or beam-off sample. The T2 negatives are overlay events in which the neutrino left no charge.

- **Beam-window light only** (23.4 μs). This is not Wire-Cell's full 4.8 ms charge–light matching.
- **Single seeds** at 10% and 100% labels and in E4/E6. Differences below about 0.02 are not significant on their own.
- **Fixed downstream budget.** At large label budgets, scratch is not trained to convergence.
- **Small models and budget.** The models (6.6M and 21M parameters) and the few-million-event pretraining budget were set by the 100 node-hour allowance.
- **Non-natural T3 class proportions.** The ν_e sample is mixed into all splits.
- **Operational label thresholds.** For example, the hit-level ν threshold of 0.05 MeV.

8 Next steps

1. **Hit-level representations.** Add a per-hit pretraining signal: hit-level masked prediction or dense self-distillation [9, 11]. Also try finer cells.
2. **Latent self-distillation objective.** An iBOT/JEPA-style objective [12] compared with MAE at matched compute.
3. **Scaling.** Grow model size and compute together, with more seeds, and add the 1% and 50% points for the M model.
4. **Backbone ablation.** FM4NPP-style per-hit tokens with a Mamba2 backbone [8].
5. **Physics realism.** Beam-off samples; interaction-level (slice) heads; a comparison with Wire-Cell's generic neutrino selection [3] at matched efficiency.

Reproducibility

- **Code:** `https://github.com/HaiwangYu/fm4npp-lartpc`, branch `round1-prototype`.
- **Main entry points:** `qlfm.data.convert`, `scripts/make_splits.py`, `qlfm.train_pretrain`, `qlfm.train_downstream`, `scripts/make_tasks.py`, `scripts/feeder.sh`, `scripts/summarize.py` and `scripts/plot_scaling.py`.
- **Outputs:** on Polaris under `/lus/eagle/projects/neutrinoGPU/yuhw/fm4npp-lartpc-work/`.
- **Companion documents:** a Markdown version of this note with the same content (`docs/technote-round1.md`) and a working log of decisions and findings (`docs/round1-log.md`).

References

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